

Challenge (Habanero)

- Q1.** A builder needs to mix $3 \cdot 10^3$ kg of cement with $2 \cdot 10^3$ kg of sand. What is the unit rate of cement to sand?
(A) 1.5 kg of cement/kg of sand
(B) 1.2 kg of cement/kg of sand
(C) 1 kg of cement/kg of sand
(D) 0.8 kg of cement/kg of sand
(E) 0.6 kg of cement/kg of sand
- Q2.** A recipe for making cookies requires $2\frac{1}{4}$ cups of flour to make $3\frac{1}{2}$ dozen cookies. How many cups of flour are needed to make 5 dozen cookies?
(A) 3 cups
(B) 3.5 cups
(C) 4 cups
(D) 4.5 cups
(E) 5 cups
- Q3.** The cost of 5 meters of wire is 12.
What is the cost per meter of wire?
(A) 2
(B) 2.20
(C) 2.40
(D) 2.50
(E) 2.80
- Q4.** A construction worker can lay 6 bricks in 2 minutes. How many bricks can the worker lay in 5 minutes?
(A) 12 bricks
(B) 13 bricks
(C) 14 bricks
(D) 15 bricks
(E) 16 bricks
- Q5.** A chef needs to cook $2\frac{1}{2}$ kg of rice for 8 people. How much rice is needed for 5 people?
(A) 1.5 kg
(B) 1.6 kg
(C) 1.7 kg
(D) 1.8 kg
(E) 2 kg
- Q6.** The distance between two buildings is 240 meters. If a person walks at a rate of 4 meters per second, how many seconds will it take to walk between the two buildings?
(A) 50 seconds
(B) 55 seconds
(C) 60 seconds
(D) 65 seconds
(E) 70 seconds
- Q7.** A group of workers can build 2 houses in 6 months. How many houses can they build in 9 months?
(A) 2.5 houses
(B) 2.7 houses
(C) 3 houses
(D) 3.2 houses
(E) 3.5 houses
- Q8.** A car travels 250 miles in 5 hours. How many miles will it travel in 7 hours?
(A) 350 miles
(B) 375 miles
(C) 400 miles
(D) 425 miles
(E) 450 miles

Open Ended Questions (FRQ)

- Q9.** A bakery sells a total of 250 loaves of bread per day. They sell a combination of whole wheat and white bread. The ratio of whole wheat to white bread is 3 : 5. If the bakery makes a profit of 0.50 per loaf of whole wheat bread and 0.75 per loaf of white bread, how much profit does the bakery make in a day?
- Q10.** A construction company is building a skyscraper that requires 15,000 cubic meters of concrete. The company has two concrete mixers that can produce 200 cubic meters of concrete per hour. However, the mixers can only be used for 8 hours per day due to noise restrictions. How many days will it take to produce the required amount of concrete?